

5. Given access to the same facts, how is it possible that there can be disagreement between experts in a discipline? Develop your answer with reference to two areas of knowledge.

In my studies in school, I have come to learn that it is possible for experts to disagree in a discipline despite having access to the same facts. It may seem unlikely but disagreement between experts occur often in areas such as law, politics but also in sciences. Facts can be defined as proven, indisputable, shared knowledge within a society while experts can be defined as members in a society who have extensive knowledge covering the discipline they specialize in. Experts have access to the same facts, when they have the same knowledge available to all of them. Theories are ideas that are made with the intention of explaining situations and are usually established from facts. However, theories could differ between experts because of the use of different reasoning processes such as deductive or inductive reasoning and having different experiences and memories. These differences could influence experts who even have access to the same facts to disagree because their way of thinking would be different and they would have each developed different personal bias from their memories and experiences. **This leads me to the following primary knowledge question, how do reasoning and memories influence experts' interpretation of facts?**

In the natural sciences, the type of reasoning used by experts can lead to different theories resulting in disagreements in the discipline. In physics, this can be demonstrated with the case of Olbers' Paradox. Heinrich Wilhelm Olbers lived in a time where the idea of an infinite universe was feasible and reasonable (Darnell, 2011). However, Olbers questioned the idea of an infinite universe, with a simple but compelling question: why is the night sky dark, if the universe is infinite (Lawrence Berkeley National Laboratory Physics Division, n.d.). Olbers' argument infused deductive reasoning by going from general to specific. If the universe was infinite, there would be a star at each point you would look at in the sky

(Lawrence Berkeley National Laboratory Physics Division, n.d.). Therefore, the sky would be bright since there would be a star in each point in the sky you would look at (Lawrence Berkeley National Laboratory Physics Division, n.d.). In reality, the sky is dark; therefore how could the universe be infinite hence creating the paradox. On the other hand, Giordano Bruno came to the conclusion that the universe was infinite through inductive reasoning. He argued that if it was possible for a person to think that another world exists then it is possible that infinite(all) worlds exist (Giorbran, n.d). It can be seen that his argument infused inductive reasoning as his argument went from specific where you imagine another world exists to general where you imagine infinite worlds exist. The disagreement between Giordano and Olber arose from the use of different types of reasoning and affirms the claim that the type of reasoning used by experts can lead experts to disagree. It can be said that these experts had access to the same facts as the fundamental basis of Olbers' paradox is the common fact that the sky is dark at night.

Although the type of reasoning influences disagreements among experts, memories and experiences may play a bigger role in influencing disagreements within experts. Thomas Digges was an expert who believed in Nicolas Copernicus' ideas of that the sun was the center of our solar system and planets orbit around it (a heliocentric solar system) but also had ideas that the universe was infinite (Ashworth, 2015). This is evident as Digges is considered to be the first Copernican in England: the first person to accept and defend the heliocentric solar system in England (Ashworth, 2015). The reason why he was able to be the first in England pioneer this theory could be because he was born very close to the time that Nicolas Copernicus lived. This is supported by the availability heuristic theory, which states that recent occurrences appear to be more probable to people (Straker, 2002). Thomas transformed the shared knowledge of Copernicus to personal knowledge by adding the idea of an infinite universe. Digges did this firstly, by drawing a diagram of a heliocentric solar

system but with stars extending in the periphery of the system “infinitely” (Ashworth, 2015) which would give the idea the universe is infinite. He also added his personal ideas of an infinite universe in his translation of Copernicus’s *De Revolutionibus* (O’Connor and Robertson, 2002). It can be argued that the memories or experiences that Thomas had from the Copernican theory provided the platform for him to have ideas of an infinite universe and he may have not arrived to his conclusion without them. In comparison, Olbers was born much further from the time Copernicus lived, therefore his memories of it may not have been strong enough to influence him in the way that it influenced Digges. This is evident in their theories as Digges’ theory is centered around Copernicus’ system while Olbers’ theory is based on the finding out the reason why the sky is dark at night.

However, these ways of knowing may have different effects in Human and Natural Sciences.

This can lead to the following subsidiary knowledge question: what influence does reasoning and memories have in human and natural sciences?

Memories and experiences have a strong influence in the natural sciences but they can also influence disagreement between experts in the human sciences as well. Paul Ehrlich who had written a book titled, “The Population Bomb” in which he predicted a famine in the 1970s that would cause a great number of people to starve to death (Codrington, 2013). Paul Ehrlich’s claim could have been based on the Malthusian theory. Malthusian theory argues that population increases exponentially while resources increase arithmetically therefore unrestricted population growth would outstrip resources (Boundless, 2016). Malthusian theory invokes deductive reasoning as it goes from general (population increases exponentially while resources increased arithmetically) to specific, population growth will outstrip resources. Julian Simon arrived at a different conclusion from Paul while still using deductive reasoning. He said price was the true the measure of scarcity and since there had been a decrease in price of resources in the past years, it would mean that resources were

becoming more abundant and not becoming scarce (Codrington,2013). This would mean that despite the increase in population, resources are becoming more abundant instead of decreasing. Even though they used the same type of reasoning, they may have arrived at different conclusion because of their experiences and memories. When growing up, Paul Ehlich may have experienced goods getting more expensive in the area where he lived while Simon may have had an opposite experience that life was getting better and cheaper to afford. This may have played a role in their theories as each may have looked for evidence to support the experience they believed was true and hence resulted in a disagreement between the two experts. This would affirm the claim that in the human sciences, memories play a bigger role than reasoning in influencing disagreements among experts who have access to the same facts.

These experts may have both utilized deductive reasoning but their reasoning on how human beings behave is very different. Experts in the human sciences can disagree based on their reasoning on how humans behave. Paul related animal behavior to human behavior in his book, “the Population Bomb” (Codrington,2013). Animals don’t actively try to replenish or increase the amount of resources they consume. For example, hen we put too many cows on a piece of land than the land can sustain the vegetation cover does not increase but it. Paul may have believed the same thing was occurring with humans on earth. On the other hand, Julian Simon believed that humans behaved differently to animals and liked to quote Henry George. “Both the Jayhawk and the man eat chickens, but the more jayhawks, the fewer chickens while the more men, the more chickens.” (Codrington,2013). Julian’s argument essentially differentiates humans from animals argues that humans find ways to replenish or increase the amount of resources they have consumed. Animals consume but don’t usually attempt to replenish or increase the amount of resources they consume. This would explain why Julian thought resources are getting more abundant while Paul believed

they were getting scarce. The disagreement between these two experts arose from their reasoning on how human beings behave. This supports the counterclaim, that in the human sciences, the reasoning and outlook on how human beings can lead to disagreement between experts even though they may have access to the same facts.

In conclusion, it can be seen that both memories and reasoning play a large role on how experts interpret facts. It is possible for experts to disagree despite having access to the same facts depending on their reasoning and memories/experiences they have. In the natural sciences, it can be seen that reasoning can play a major role in how experts interpret facts with the case of Olbers and Giordano Bruno as they disagreed because they each used a different type of reasoning. In the human sciences, it can be seen that the experts disagreed based on their reasoning on how humans behave. Memories can also have influence how experts interpret facts in the human and natural sciences and lead to disagreement as seen Digges and Olbers in Natural Sciences and Julian Paul in Human Sciences. Therefore, reasoning and memories can influence how experts interpret facts and make it possible for them to agree or disagree.

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